

# Rahil Piyush Mehta

REST API Testing Researcher · ICSE 2026 First Author · Backend Systems

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## Education

**Oregon State University**, *Master of Science in Computer Science*

Sept 2023 – June 2026

**Mumbai University**, *Bachelor of Engineering*

May 2018 – May 2022

## Experience

**Graduate Research Assistant**, *Oregon State University*

May 2024 – April 2026

- Built **Defects4REST** (ICSE'26 Research Track), a benchmark of **110 real-world REST API defects** across 12 containerized services with single-command Docker reproducibility; accepted to ICSE 2026, top-tier software engineering venues (~**20% acceptance rate**).
- Ran **600+ hours** of HPC testing campaigns evaluating **Schemathesis**, **RESTler**, **EvoMaster**, and **AutoRestTest** across all defects; mined logs, HAR traces, and test reports to quantify tool effectiveness revealing a ~**90% defect miss rate** across all evaluated tools.
- Built an environment generation engine in **C#** (AAAI'25) that systematically samples diverse test configurations for autonomous systems, enabling **differential testing** that surfaced latent bugs undetectable through manual testing.
- Designed a **modified beam search** in Python to assess task feasibility across generated environments, automating the detection of systemic errors in autonomous agent configurations at scale.

**Graduate Teaching Assistant**, *(CS 362: Software Testing) Oregon State University*

Sept 2024 – April 2026

- Graded mutation testing and coverage assignments for **100+ students** across 3 semesters; mentored on **CI/CD**, code reviews, and debugging during weekly office hours.

**Software Engineer**, *Nerve Solutions (Startup)*

Feb 2023 – Aug 2023

- Designed and shipped a high-throughput **.NET alert engine** integrated with **MongoDB**, processing **250K+ alerts/second**, incorporating user preferences, alert priority, and real-time system load.
- Cut infrastructure costs by **33%** by replacing legacy alert routing logic with a CPU and memory-efficient algorithm without degrading throughput or reliability.
- Increased system resilience by adding **load balancing**, **Caching**, structured **logging**, and **automatic failover**; reduced unplanned downtime and improved observability across the alert pipeline.

**Programmer**, *Acty System India (Subsidiary of Acty System Japan)*

May 2022 – Nov 2022

- Built a **WPF** desktop app for real-time, flicker-free 2D graphing using **multi-threading** and double buffering — no external libraries.
- Implemented login, password reset, and CRUD modules for an internal **Employee Management System** used by HR and admin staff.

## Projects

**ScrollTone – Self-Hosted Audiobook Generator (2024)**

- Converts EPUB files to audiobooks using **Kokoro-82M TTS** with 19 voices and multi-character speaker attribution via a local **Ollama** LLM; fully **Dockerized** with a **FastAPI** backend, no cloud APIs, no accounts, no recurring cost.
- Supports per-chapter voice assignment and batch processing; built as a personal tool, now used as part of a self-hosted home media stack.

**mornin-cli – Git-to-Standup CLI Tool (2026)**

- CLI tool (pow) that generates daily standup summaries from **git commit history** using a local **Ollama** LLM; degrades gracefully to a rule-based fallback when no LLM is available.
- Single command, zero dependencies beyond git — outputs plain text ready to paste into Slack or email.

## Publications

- Rahil Piyush Mehta** et al., “Defects4REST: A Benchmark of Real-World Defects to Enable Controlled Testing and Debugging Studies for REST APIs.” *ICSE 2026 (Research Track)*. **First Author. Role:** Led benchmark design/implementation, built deployment + replication tooling, and ran empirical evaluation.
- Rahil Piyush Mehta** et al., “Uncovering Systemic and Environment Errors in Autonomous Systems Using Differential Testing.” *AAAI Fall Symposium 2025 (ATRACC)*. **Co-First Author. Role:** Built env/task generation, feasibility detection, and experimental analysis.

## Open Source Contributions

- Schemathesis** – Reported #3377: `KeyError: 'operationId'` crash in stateful link resolution on OpenAPI 3.1.0 specs. *Fixed*
- AutoRestTest** – Reported 4 bugs: #34 `KeyError: 'application/xml'` crash (*fixed*); #36 graph init hang on large specs (*acknowledged*); #31 value table stuck on Podman spec (*acknowledged*); #30 missing custom API key header support (*feature added*).
- EvoMaster** – Reported #1400: `IllegalArgumentException: invalid start or end` crash during Dolibarr API testing. *Fixed*.

## Skills

**Languages:** Python, C#, Swift, SQL    **Frameworks & Tools:** .NET, FastAPI, Docker, GitHub Actions, Bash, Linux, Git, HPC/SLURM    **Dominant:** REST APIs, OpenAPI/Swagger, API Fuzzing, Ollama, SwiftUI